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Internship Report — SI-26 Cohort

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INTERN NAME Humna Imran	**STUDENT ID** 2023-BS-AI-017
DEPARTMENT ML / AI	**REPORT NUMBER** Report 1 (Week 1–3)
HOST INSTITUTION Code Saviours (SMC-PRIVATE) Limited	**REPORTING PERIOD** 29/06/2026 – 17/07/2026
FACULTY SUPERVISOR —	**SITE SUPERVISOR** Code Saviours (SMC-PRIVATE) Limited

A. Tasks performed

During the first three weeks, my primary duty was to lay the groundwork for training a custom Urdu OCR model. Specifically, I completed the following:

Week 1 (Setup & Collection): Established the development environment by setting up Google Colab, GitHub, and Hugging Face. I collected, organized, and labeled an initial dataset of 100+ Urdu images, structuring them into a repository with a corresponding labels.csv file.

Week 2 (Preprocessing & Gap Analysis): Engineered an image preprocessing pipeline utilizing OpenCV (applying techniques like grayscaling and thresholding) to clean the raw image data. I then ran baseline text extraction tests using Tesseract OCR and documented a gap analysis detailing its high failure rate on the Urdu script.

Week 3 (Dataset Expansion & Data Loader): Expanded the foundational dataset to include over 200 image-text pairs. I then developed the core data pipeline by writing a custom PyTorch Dataset class and integrating Hugging Face processors, ensuring the image and text data batches load perfectly into the TrOCR architecture.

B. Learning experience

Over this period, I significantly improved my practical skills in computer vision and data engineering for machine learning. I gained hands-on experience using OpenCV for image manipulation and PyTorch for constructing custom data loaders. Conceptually, conducting the gap analysis on Tesseract taught me how to evaluate model baselines and identify specific algorithmic limitations. Most importantly, building the data pipeline from scratch changed how I approach AI projects; I now have a much deeper appreciation for the fact that a model can only be as effective as the quality and structure of the data it learns from, teaching me to prioritize meticulous data cleaning and loading before attempting any model training.

C. Challenges and how handled

Challenge: The most notable challenge was observing the highly inaccurate outputs from existing tools like Tesseract when processing the cursive Urdu script.

Handling: Rather than trying to force an incompatible tool to work, I used this as an analytical opportunity. I meticulously documented Tesseract's specific character recognition failures to establish a clear gap analysis, proving the necessity of fine-tuning our own TrOCR model.

Challenge: Bridging the gap between raw CSV/image folders and a format Hugging Face models can digest was technically demanding.

Handling: I tackled this by heavily researching PyTorch Dataset documentation and systematically debugging my data loader until it successfully iterated through batches.

Future Approach: Going forward, I would dedicate more upfront time to sourcing a highly diverse set of images from the start, as expanding the dataset in Week 3 was more time-consuming than anticipated.

Intern signature & date

Supervisor signature & date